

BERGA, Sweden

WMO: 024890

Lat: 59.069N

Lon: 18.118E

Elev: 4

StdP: 101.28

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.2	-10.9	-16.3	0.9	-13.6	-13.3	1.2	-9.7	10.3	1.2	9.2	1.7	1.6	270	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	25.0	19.0	23.3	18.2	21.9	17.2	20.3	23.5	19.2	22.1	18.2	20.7	3.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	13.9	21.8	18.1	13.0	20.5	17.1	12.2	19.7	58.2	23.6	54.6	22.1	51.4	20.6	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 9.0	7.6	6.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			-17.5	21.2	5.4	1.5	-21.4	22.3	-24.6	23.2	-27.6	24.0	-31.5	25.1	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-1.4	-1.4	0.9	4.7	9.9	14.2	17.3	16.4	12.4	7.1	3.4	-0.2	(6)	
	DBStd	7.49	3.90	4.42	3.93	3.11	3.32	2.57	2.52	2.14	2.88	3.41	3.35	4.56	(7)	
	HDD10.0	1786	353	319	282	161	42	1	0	0	12	101	199	316	(8)	
	HDD18.3	4161	611	553	540	409	261	124	49	66	178	347	448	574	(9)	
	CDD10.0	694	0	0	1	2	40	128	228	199	84	12	1	0	(10)	
	CDD18.3	27	0	0	0	0	0	1	19	7	1	0	0	0	(11)	
	CDH23.3	147	0	0	0	0	1	9	115	21	1	0	0	0	(12)	
	CDH26.7	13	0	0	0	0	0	0	13	1	0	0	0	0	(13)	
(14) Wind	WSAvg	3.2	3.4	3.4	3.2	3.0	2.9	3.0	3.0	2.8	3.0	3.3	3.5	3.6	(14)	
(15) Precipitation	PrecAvg	601	44	35	32	35	39	61	60	66	59	58	58	56	(15)	
	PrecMax	679	89	75	77	89	96	107	114	153	168	125	124	133	(16)	
	PrecMin	411	6	8	5	6	14	16	16	10	7	8	11	11	(17)	
	PrecStd	59	22	14	16	23	19	29	30	40	37	26	28	28	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.3	8.5	14.0	16.6	22.4	24.7	28.2	25.4	21.7	15.9	11.5	9.8	(19)	
		MCWB	5.7	5.4	8.6	10.2	15.0	17.2	20.4	19.9	16.5	12.7	10.3	8.3	(20)	
	2%	DB	5.8	6.6	10.6	14.0	19.8	22.5	26.2	23.6	19.8	14.2	10.1	7.9	(21)	
		MCWB	4.5	4.6	6.5	8.7	13.8	16.1	20.0	18.8	16.0	12.2	9.2	6.7	(22)	
	5%	DB	4.5	5.2	8.4	12.0	17.8	21.0	24.4	22.2	18.4	13.1	9.0	6.5	(23)	
		MCWB	3.6	3.6	5.0	7.6	12.3	15.7	18.9	17.9	15.2	11.6	8.1	5.6	(24)	
	10%	DB	3.4	3.8	6.3	10.1	16.0	19.5	22.8	21.0	17.2	12.3	8.0	5.5	(25)	
		MCWB	2.6	2.8	3.8	6.4	11.5	15.0	18.3	17.2	14.5	11.0	7.2	4.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	6.2	9.0	11.1	16.1	18.6	22.0	21.5	18.0	13.7	10.7	8.5	(27)	
		MCDB	6.9	7.5	13.2	15.3	20.9	22.1	25.7	24.1	20.1	14.7	11.0	9.4	(28)	
	2%	WB	4.7	4.9	6.8	9.2	14.4	17.4	20.8	19.8	16.6	12.7	9.4	6.8	(29)	
		MCDB	5.5	6.4	10.0	13.5	18.4	20.8	24.7	22.2	18.9	13.7	10.1	7.4	(30)	
	5%	WB	3.7	3.8	5.4	8.0	13.1	16.4	19.9	18.8	15.7	12.0	8.2	5.8	(31)	
		MCDB	4.4	4.9	8.0	11.1	16.8	19.6	23.3	21.0	17.5	12.9	8.8	6.3	(32)	
	10%	WB	2.7	3.0	4.1	6.9	12.0	15.6	18.8	18.0	15.0	11.1	7.3	4.8	(33)	
		MCDB	3.3	3.8	5.8	9.5	15.3	18.5	22.0	20.0	16.7	12.0	7.9	5.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	5.4	6.8	8.5	8.4	8.2	8.2	7.9	7.2	5.7	4.3	4.3	(35)	
		MCDBR	5.0	6.2	9.5	11.4	11.7	10.8	10.7	9.6	8.9	6.3	5.6	4.4	(36)	
	5% WB	MCWBR	4.7	4.6	6.1	6.9	6.6	5.7	5.5	5.4	4.5	5.1	4.2	4.2	(37)	
		MCWBR	5.2	5.3	9.0	10.9	10.6	9.3	9.7	7.9	7.5	5.2	5.1	4.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.289	0.325	0.342	0.379	0.375	0.365	0.394	0.384	0.352	0.340	0.316	0.275	(40)	
		taud	2.231	2.293	2.353	2.318	2.400	2.464	2.404	2.422	2.495	2.412	2.258	2.154	(41)	
	Ebn at Noon		520	687	796	823	856	870	832	812	785	667	474	390	(42)	
		Edh at Noon	44	70	87	107	106	101	105	96	76	62	44	33	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.30	0.86	2.39	4.06	5.43	6.00	5.55	4.39	2.85	1.24	0.40	0.20	(44)	
	RadStd		0.04	0.15	0.25	0.38	0.44	0.32	0.51	0.37	0.31	0.20	0.09	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	+0.37	N/A	N/A	N/A	N/A	N/A	-116	-145	N/A	N/A

Nomenclature: See separate page